

INTERNSHIP REPORT

ON

DATA ANALYST

Submitted in Partial Fulfilment

For the Award of the Degree of

Bachelor of Technology

COMPUTER SCIENCE



B.TECH CSE

(SECTION-B)

[Internship at Atharav Tech Solutions LLP.](#)

[Role: Data Analyst \(Intern\): 15/09/2025 - 29/04/2026](#)

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

GAUTAM BUDDHA UNIVERSITY

GREATER NOIDA –201312, GAUTAM BUDDHA

NAGAR, UTTAR PRADESH , INDIA

Submitted to (Supervisor)

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Submitted By-

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225/UCF/049

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **AtharavTech Solutions LLP** for providing me the opportunity to work as a Data Analyst Intern. This internship has been an invaluable learning experience and has allowed me to gain practical exposure in real-time drone-based data analytics.

I am extremely thankful to my internship supervisor, **Ms. SANGEETA**, for his continuous guidance, feedback, and support throughout the training period. His insights and encouragement helped me complete my assigned tasks effectively.

I also extend my gratitude to the faculty members of **Gautam Buddha University**, especially the School of Information and Communication Technology, for providing the support and academic foundation required for this project.

Lastly, I thank my family and friends for their support and motivation during this internship journey.

DECLARATION

I, **Harshit Rathore**, Roll Number **225/UCF/049**, hereby declare that the report entitled: **“Industrial Training Report on Data Analytics in Drone Technology”**

is a result of my own work carried out at **AtharavTech Solutions LLP (Drone Technology Division)** from **15 September 2025 to 29 April 2026** as part of the B.Tech (Computer Science) program at Gautam Buddha University.

This report has not been submitted to any other institution for the award of any degree or diploma.

(Harshit Rathore)

Roll No: 225/UCF/049



School of Information & Communication Technology

GAUTAM BUDDHA UNIVERSITY

GREATER NOIDA

CERTIFICATE

This is to certify that **Harshit Rathore (Roll No: 225/UCF/049)** has successfully completed his internship as a **Data Analyst Intern** at **AtharavTech Solutions LLP (Drone Technology Division)** from **15 September 2025 to 29 April 2026**.

His performance during the internship period was found satisfactory, and this report is accepted as a part of the Industrial Training requirements for the award of **B.Tech (Computer Science)**.

Ms. SANGEETA
(ASSISTANT PROFESSOR)

ABSTRACT

This industrial training report summarizes the six-month internship experience completed at **AtharavTech Solutions LLP** in the Drone Technology Division. The focus of the internship was on working with drone-generated data including flight telemetry, GPS logs, aerial imagery, and performance metrics.

The internship involved data collection, cleaning, preprocessing, exploratory analysis, dashboard visualization, and analytics support for object detection and predictive maintenance. Tools such as **Python, Pandas, Power BI, SQL, and GIS mapping tools** were used during the internship.

The report highlights key tasks, software used, challenges faced, and the knowledge gained. The overall internship experience improved my technical skills, analytical thinking, problem-solving mindset, and understanding of data-driven decision processes in drone analytics.

CHAPTER 1 — INTRODUCTION

1.1 About Data Analytics in Drone Technology

Drone analytics involves extracting meaningful insights from aerial and telemetry data collected by unmanned aerial vehicles (UAVs). Modern drones generate large datasets including:

- GPS coordinates
- Battery consumption logs
- Motor performance data
- Thermal, RGB or LiDAR imaging
- Object detection and surveillance video feeds

This data helps industries optimize operations in:

Industry	Use-Case
Agriculture	Crop health monitoring, yield estimation
Logistics	Route optimization and delivery path planning
Surveillance	Object tracking and border monitoring
Contruction	Site mapping, measurement, and inspection

1.2 Background

With increasing commercial use of drones, there is a growing need for efficient data storage, processing, visualization, and predictive insights. Companies rely on data analysts to transform raw flight data into actionable metrics.

1.3 Objective

- To analyze real-time drone telemetry and imagery datasets
- To build dashboards for operational decision-making
- To support predictive maintenance using performance trends
- To prepare structured datasets for AI/ML models

1.4 Scope

The internship covered:

- Data collection & cleaning
- Data processing using Python
- Dashboard designing using Power BI
- SQL-based Data Reporting
- Flight Telemetry Trend Analysis
- Machine Learning Data Preparation

CHAPTER 2 — TASKS PERFORMED

Task 1: Drone Data Collection & Cleaning

Tools Used Python, Pandas, Jupyter Notebook, GPSTools

Extracted flight logs and converted raw drone telemetry (GPS, altitude, battery status) into structured tabular format.

Removed noise, inconsistencies, missing values.

Standardized units (meters, latitude/longitude, timestamps).

Task 2: Exploratory Data Analysis

Tools Python, Matplotlib, Seaborn

Performed:

- Trend analysis of battery drain over flight duration
- Path efficiency analysis
- Speed vs altitude correlation
- Distance traveled vs power usage patterns

Generated summary reports and visual insights.

Task 3: Dashboard Creation for Business Insights

| Tool Used | Microsoft Power BI |

Built dashboards showing:

- Mission success rate
- Fleet performance
- Battery efficiency trends
- Live operational status

Task 4: Predictive Maintenance Analysis

Used historical drone logs to:

- Identify anomaly patterns
- Predict hardware failure risks
- Support maintenance scheduling

Task 5: Image Dataset Labeling for ML

Assisted AI team by:

- Labeling and annotating objects from aerial imagery
- Organizing structured datasets for training models

CHAPTER 3 — CHALLENGES FACED

During the internship at **AtharavTech Solutions LLP (Drone Technology Division)**, several technical and operational challenges were encountered while working with large-scale drone datasets and analytics workflows. These challenges became learning opportunities and helped in strengthening both technical proficiency and analytical skills.

3.1 Handling Inconsistent Sensor Formats

Drone data was collected from multiple drone models and firmware versions, leading to

- File formats (JSON, CSV, TXT, GPX, BIN)
- Log structure (missing or additional parameters)
- Units (imperial vs metric)
- Timestamp formats and GPS precision

This required designing a **standardized preprocessing workflow** to normalize data fields before analysis. Tools like **Python Pandas, GPSBabel, and custom scripts** were used to convert, align, and validate different data structures.

3.2 Large Data Size Processing

Drone-generated flight logs and aerial imagery often reached sizes ranging from a few hundred MBs to several GBs. Processing these files on a standard system initially resulted in:

- Long computation times
- RAM limitations
- Crashing of analytical notebooks

To overcome this, techniques such as:

- **Chunk loading**
- **Lazy evaluation using Dask**
- **Optimized Pandas operations**
- **Using compressed file formats (Parquet)**

were implemented to ensure smooth and efficient data handling.

3.3 Integrating Telemetry with Geospatial Tools

Mapping drone flight paths and overlaying telemetry values on a geospatial layer required working with GIS tools and coordinate transformation libraries. Challenges included:

- Converting GPS logs into shapefile or GeoJSON formats
- Handling coordinate reference systems
- Plotting drone routes with elevation, speed, or battery overlays

Tools such as **QGIS, GeoPandas, Leaflet maps, and GIS plugins** were explored to build accurate spatial visualizations.

3.4 Learning Advanced Power BI Features

While basic dashboard creation was straightforward, applying advanced BI functionality required additional learning and experimentation with:

- DAX formulas
- Custom visuals and drill-down functionality
- Live data refresh pipelines
- Creating interactive reports with filters and conditional formatting

This challenge helped improve dashboard design skills and taught the importance of building user-friendly analytics visualizations.

3.5 Noise and Missing Values in Sensor Data

Drone telemetry often contained:

- Sudden spikes in altitude or speed values
- Missing timestamps
- Corrupted values due to signal loss

To address this, multiple data cleaning techniques were applied such as:

- Outlier detection
- Forward/backward fill methods
- Interpolation of missing coordinates
- Smoothing filters for signal noise reduction

3.6 Aligning Multidisciplinary Knowledge

Drone analytics involves a combination of:

- **Aviation parameters**
- **Statistics**
- **Programming**
- **GIS**
- **Machine Learning concepts**

Understanding how these domains interact within a real-world workflow required time, research, and mentorship guidance.

3.7 Working with Real-Time Data Constraints

Unlike classroom datasets, real-world drone data is:

- Unpredictable
- Continuously expanding
- Dependent on field operations and mission schedules

This required maintaining **version control, structured folder systems, and a scalable pipeline** to handle new incoming data efficiently.

3.8 Debugging Automation Scripts

Automation scripts failed occasionally due to:

- Data format changes
- Missing field names
- API authentication issues
- System compatibility challenges

This helped strengthen debugging skills and understanding of error-handling strategies

CHAPTER 4 — OUTCOME

The internship at **AtharavTech Solutions LLP (Drone Technology Division)** resulted in significant technical growth, professional development, and hands-on familiarity with real-time drone analytics workflows. The experience bridged the gap between theoretical knowledge and industry implementation, enabling practical application of data analysis tools, domain concepts, and analytical decision-making.

4.1 Strengthened Technical Skillset

Throughout the internship, programming and analytical capabilities improved significantly. The use of **Python, SQL, Pandas, Power BI, and geospatial libraries** enabled exposure to advanced data processing workflows and automated analytics pipelines. Working with real datasets strengthened the ability to write efficient, optimized, and reusable code.

Key technical improvements include:

- Writing Python scripts for cleaning and transforming telemetry logs
- Creating SQL queries for data retrieval and reporting
- Understanding modular code structure for scalable analytics
- Using data visualization to uncover meaningful operational trends

4.2 Hands-on Experience with Drone Data Lifecycle

The internship provided a full understanding of the **end-to-end drone data workflow**, from raw data acquisition to the delivery of client-ready dashboards. Exposure to this pipeline allowed for practical familiarity with:

- Telemetry logging
- GPS data processing
- Geospatial visualization
- Statistical trend analysis
- Performance interpretation

This hands-on exposure helped develop accuracy and confidence while handling live drone-based datasets.

4.3 Improved Understanding of Real-World Data Engineering

Working with large, unstructured datasets introduced concepts beyond normal coursework, including:

- Data storage formats (Parquet, JSON logs, GPX files)
- Efficiency-based processing (chunking, lazy loading, caching)
- Error handling for corrupted or incomplete flight logs
- Standardization of multiple sensor streams

These skills contribute to better understanding of how data engineering supports analytics and machine learning environments.

4.4 Analytical Thinking and Problem-Solving Growth

Throughout the internship, analytical reasoning skills improved through:

- Identifying insightful patterns in drone-captured data
- Interpreting anomalies in speed, altitude, or power usage
- Comparing multiple mission metrics for optimization
- Evaluating the impact of flight conditions on performance

This enhanced the ability to convert raw data into strategic insights.

4.5 Dashboard and Reporting Competency

One of the major outcomes was the ability to design and deploy interactive dashboards using **Power BI**, supporting operations and decision-making. The dashboards included:

- Fleet performance overview
- Mission success and failure trends
- Battery consumption and predictive maintenance analysis
- Geospatial drone flight visualization

These dashboards simplified complex datasets into a **business-friendly** reporting interface.

4.6 Professional & Workplace Learning

Beyond technical skills, the internship contributed to essential professional development, including:

- Working in a structured corporate environment
- Participating in technical discussions and review meetings
- Managing time, deadlines, and task ownership
- Documenting workflows and maintaining organized datasets
- Following company data security and compliance protocols

These experiences strengthened communication, teamwork, and accountability.

4.7 Contribution to Live Use Cases

The work completed during this internship contributed to:

- Improving fleet monitoring accuracy
- Enhancing analytics for predictive maintenance
- Supporting dataset preparation for AI-based automation
- Streamlining data reporting workflow for business operations

This created real measurable value for the organization.

Task 1: Drone Data Collection & Cleaning

This task involved acquiring, organizing, and standardizing drone-generated telemetry datasets to prepare them for deeper analysis and dashboard visualization. The primary objective of this stage was to ensure that raw sensor data was converted into a reliable, structured, and analysis-ready format.

| **Tools Used** | **Python, Pandas, NumPy, Jupyter Notebook, GPSTools, Visual Studio Code, CSV/JSON Parsers** |

1. Data Acquisition

Raw datasets were obtained from:

- Drone onboard storage systems
- Cloud-synced flight logs
- Mission control platforms
- GPS and telemetry export files
- Camera metadata and sensor logs

These datasets were initially available in multiple formats such as **.CSV, .JSON, .GPX, .LOG, and .TXT**, depending on drone hardware and firmware versions.

2. Data Parsing and Format Conversion

Since different drone models used varying data structures, tools such as **GPSTools** and Python scripts were used to convert raw formats into a uniform structured format. Multiple conversions included:

- **GPX → CSV**
- **JSON → Tabular DataFrame**
- **Raw log files → Parsed telemetry records**

This ensured standardization for further automation and analysis.

3. Data Cleaning and Filtering

The collected data contained noise due to real-world environmental variations (signal loss, altitude fluctuations, duplicated timestamps, etc.). The following steps were performed:

- Removal of **duplicate records**
- Handling **missing GPS coordinates** with interpolation
- Correction of **timestamp alignment and format mismatch**
- Filtering **false spikes in speed, altitude, and acceleration values**

Advanced techniques such as **rolling mean filters, interpolation, logical constraints, and smoothing algorithms** were applied to stabilize noisy sensor values.

4. Data Standardization

To ensure consistency across datasets, normalization rules were applied including:

Parameter	Standardization
Distance and coordinates	Converted into decimal latitude/longitude
Speed & Acceleration	Unified into m/s
Altitude	Standardized into meters above ground level
Time	Converted into ISO 8601 timestamp format

This step ensured that data from multiple drone models could be compared and analyzed uniformly.

5. Data Structuring & Storage

After cleaning, the final dataset was stored in structured formats such as:

- **CSV for lightweight analysis**
- **Parquet format for optimized storage**
- **SQLite / SQL tables for query-based reporting**

Metadata such as **flight ID, date, drone model, operator details, and mission purpose** were appended to maintain traceability and compliance.

6. Verification and Validation

Validation steps included:

- Visual inspection of flight paths on maps
- Random sampling checks to confirm absence of corrupted data
- Cross-referencing with recorded mission logs

This ensured that dataset integrity was maintained before analytical modeling.

Outcome of Task 1

After successful completion of this phase, the dataset became:

- Clean
- Organized
- Consistent
- Machine-learning ready
- Directly usable for analysis dashboards and predictive modeling

Task 2: Exploratory Data Analysis (EDA)

Once the cleaned and standardized drone telemetry dataset was prepared, the next step involved performing **Exploratory Data Analysis (EDA)** to understand patterns, relationships, and operational insights from the data. This analysis helped uncover meaningful trends related to drone performance, energy consumption, flight dynamics, and mission efficiency.

| **Tools Used** | Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook |

1. Statistical Summary and Initial Inspection

The EDA process began by generating descriptive statistics to understand the distribution of key variables such as:

- Flight duration
- Average altitude
- Maximum and minimum speed
- Battery percentage usage
- Distance covered

Python functions like `.describe()`, `.value_counts()`, and `.info()` were used to detect abnormalities, outliers, or unexpected values.

2. Battery Performance and Power Consumption Trends

A key focus was analyzing how battery levels changed throughout each mission. Insights included:

- Rate of battery drain over time
- Impact of altitude and speed on energy efficiency
- Variation in battery performance across drone models and missions

Visualizations such as **line graphs and area charts** helped identify consistent behavior patterns and anomalies indicating potential battery health degradation.

3. Flight Path and Route Efficiency Analysis

Using telemetry and GPS logs, flight path analysis evaluated:

- Whether drones followed optimal planned routes
- Frequency of hovering or idle time
- Impact of terrain and distance on route deviation

Scatter plots and geospatial charts helped correlate **actual route vs expected path** performance.

Compared Variables	Purpose
Speed vs Altitude	Identify aerodynamic efficiency patterns
Distance vs Battery Usage	Measure power intensity of missions
Acceleration Vs Motorload	Detect Possible Hardware stress Patterns

Heatmaps, pair plots, and regression lines helped visualize strong or weak correlations.

5. Trend and Pattern Identification

Time-series analysis techniques were used to detect:

- Consistent operational trends
- Sudden performance spikes
- Repeated battery stress points
- Environmental influence on flight efficiency

Rolling averages and smoothing filters further clarified seasonal and operational patterns across missions.

6. Visualization & Reporting

The final stage of EDA involved compiling the findings into:

- Line graphs
- Histograms
- Bar charts
- Correlation heatmaps
- Performance dashboards

These visualizations transformed raw metrics into **actionable business insights**, allowing teams to improve drone performance, plan maintenance schedules, and optimize mission execution strategies.

Outcome of Task 2

The EDA phase resulted in:

- Clear visibility of drone performance metrics
- Identification of patterns affecting battery and flight efficiency
- Insightful correlations supporting predictive maintenance
- Visualization-ready datasets for Power BI dashboards and future AI model training

Task 3: Dashboard Creation for Business Insights

The next major phase of the internship involved designing interactive dashboards to convert analytical results into visually interpretable business insights. Using **Microsoft Power BI**, multiple dashboards were developed to monitor drone fleet performance, operational safety, energy efficiency, and mission execution accuracy in real time. The dashboards served as decision-support tools for management, technical teams, and operational analysts.

| **Tool Used** | Microsoft Power BI, Power Query, DAX (Data Analysis Expressions), Data Modeler |

1. Data Import and Model Setup

The cleaned drone data from earlier stages was imported into Power BI using **Power Query Editor**. Data modeling included:

- Establishing relationships between telemetry tables
- Creating logical dimension and fact tables
- Applying star-schema modeling for optimized analytics
- Configuring refresh schedules for updated flight logs

This ensured seamless integration between raw telemetry, metadata, and analytical components.

2. Data Processing and Calculated Measures

Power BI functions and DAX formulas were used to compute advanced metrics such as:

- Average battery drop per minute
- Energy consumption rate per kilometer
- Hover vs travel time percentage
- Real-time flight status classification
- Mission reliability index

These calculated measures supported deeper operational understanding beyond raw data.

3. Dashboard Layout and UI/UX Design

To ensure clarity and ease of interpretation, dashboards included visual elements such as:

- Line charts showing **battery performance trends**
- KPI cards displaying **mission success rate**
- Pie/Donut charts for **mission type distribution**
- Maps visualizing **flight paths and geographic coverage**
- Heatmaps showing **speed vs altitude correlation**
- Time-series charts for **maintenance prediction**

Charts were positioned based on visual hierarchy, ensuring a clear and intuitive flow.

4. Real-Time Monitoring and Data Refresh

Where applicable, dashboards were linked to live or near-real-time update streams. Scheduled refresh configurations enabled:

- Hourly / daily data sync cycles
- Automatic alert triggering when thresholds exceeded (e.g., low battery

patterns) This enabled operational teams to monitor drone performance proactively.

5. Role-Based Views and Filters

Interactive filtering options were added for users to customize reports based on operational relevance, including:

- Drone model
- Date and flight duration
- Mission category
- Weather condition
- Operator
- Location coverage

Drill-down navigation allowed users to move from **fleet-level overview** to **individual drone performance analytics**.

6. Report Export and Sharing

Once finalized, dashboards were made accessible in multiple formats:

- Power BI Service cloud deployment
- PDF report export for management review
- Embedded dashboard for internal applications

Permissions were configured to protect confidential flight and operational data.

Outcome of Task 3

This task successfully produced a set of clear, interactive dashboards that improved the ability to:

- Monitor drone fleet performance
- Track operational efficiency and mission success
- Detect anomalies and hardware stress indicators early
- Support predictive maintenance planning
- Make data-driven operational decisions

The dashboards were later used in internal reporting pipelines and played a crucial role in transforming raw telemetry into meaningful, actionable insights

Project 1: Customer Churn Prediction

STEP 1 – Folder Setup (ONE TIME)

1. You already have:
`Customer-Churn-Prediction/`
2. Inside it, make: `Customer-Churn-Prediction/`
3. `data/`
4. `notebooks/`

STEP 2 – Download Dataset

1. Go to Kaggle and search: “Telco Customer Churn”
2. Download the CSV file (usually
`WA_Fn-UseC_-Telco-Customer-Churn.csv`)
3. Move it to:

`Customer-Churn-Prediction/data/`

STEP 3 – Open Jupyter Notebook & Create File

1. Open **Anaconda Navigator**
2. Click **Jupyter Notebook**
3. In the browser, go to your folder:
`Customer-Churn-Prediction/notebooks/`
4. Click **New** → **Python 3**
5. Save it as:

`churn_prediction.ipynb`

STEP 4 – Load the Dataset

`import pandas as pd`

`import numpy as np`

`import seaborn as sns`

`import matplotlib.pyplot as plt`

`%matplotlib inline`

`# Load dataset`

`df = pd.read_csv("../data/telco_churn.csv") # change name if`

`needed df.head()`

STEP 5 – Understand the Data

```
df.info()
df.describe(include='all')
df.isna().sum()
```

This tells you:

- Columns
- Types
- Missing values

Add a **Markdown cell** (click +, change to Markdown) and

write: `### Step 1: Data Loaded`

- Dataset describes telecom customers.
- We will predict 'Churn' (whether customer leaves).

(This looks professional later on GitHub.)

STEP 6 – Basic EDA (Exploratory Data

Analysis) Cell 3: Churn count

```
sns.countplot(x="Churn", data=df)
plt.title("Churn Distribution")
plt.show()
```

```
df["Churn"].value_counts(normalize=True) * 100
```

Cell 4: Relationships

```
sns.countplot(x="Contract", hue="Churn", data=df)
plt.title("Churn by Contract Type")
plt.xticks(rotation=45)
plt.show()
```

```
sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Churn by Internet Service")
plt.xticks(rotation=45)
plt.show()
```

STEP 7 – Data Cleaning

Cell 5: Fix TotalCharges & drop ID

```
# Convert TotalCharges to numeric
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

# Check missing again
df.isna().sum()

# Drop rows where TotalCharges is NaN
df = df.dropna(subset=["TotalCharges"])

# Drop customerID (not useful as feature)
df = df.drop(columns=["customerID"])

# Convert Churn to 0/1
df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})

df["Churn"].value_counts()
```

STEP 8 – Encode Categorical Columns

Cell 6:

```
# Identify categorical and numeric columns
cat_cols = df.select_dtypes(include=["object"]).columns
num_cols = df.select_dtypes(include=["int64",
"float64"]).columns

print("Categorical columns:", list(cat_cols))
print("Numeric columns:", list(num_cols))
```

Cell 7: One-hot encoding

```
df_encoded = pd.get_dummies(df, columns=cat_cols,
drop_first=True) df_encoded.head()
```

Now all features are numeric.

STEP 9 – Train/Test Split

Cell 8:

```
from sklearn.model_selection import train_test_split
```

```
X = df_encoded.drop("Churn", axis=1)
```

```
y = df_encoded["Churn"]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42, stratify=y  
)
```

```
X_train.shape, X_test.shape
```

STEP 10 – Handle Class Imbalance with

SMOTE Cell 9:

```
from imblearn.over_sampling import SMOTE  
  
smote = SMOTE(random_state=42)  
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)  
  
print("Before SMOTE:")  
print(y_train.value_counts())  
print("\nAfter SMOTE:")  
print(y_train_res.value_counts())
```

STEP 11 – Train Models

We'll use: **Logistic Regression, Random Forest,**

XGBoost Cell 10:

```
from sklearn.linear_model import LogisticRegression  
from sklearn.ensemble import RandomForestClassifier  
from xgboost import XGBClassifier  
  
log_reg = LogisticRegression(max_iter=1000)  
rf = RandomForestClassifier(random_state=42)  
xgb = XGBClassifier(random_state=42, eval_metric="logloss")  
  
models = {  
    "Logistic Regression": log_reg,  
    "Random Forest": rf,  
    "XGBoost": xgb  
}  
for name, model in models.items():  
    model.fit(X_train_res, y_train_res)  
    print(name, "trained.")
```

STEP 12 – Evaluate Models (Accuracy, Recall,

ROC-AUC) Cell 11:

```
from sklearn.metrics import accuracy_score, recall_score, roc_auc_score, classification_report
```

```
for name, model in models.items():
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[: , 1]

    print("==== ", name, " ===")
    print("Accuracy :", round(accuracy_score(y_test, y_pred), 3))
    print("Recall :", round(recall_score(y_test, y_pred), 3))
    print("ROC-AUC :", round(roc_auc_score(y_test, y_prob), 3))
    print(classification_report(y_test, y_pred))
    print("-" * 40)
```

Choose the **best model** (normally Random Forest or XGBoost) based on **highest ROC-AUC and good recall**.

STEP 13 – Plot ROC Curve for Best Model

Assume XGBoost is best. If not, replace xgb with rf etc.

Cell 12:

```
from sklearn.metrics import roc_curve

best_model = xgb # or rf / log_reg

y_prob = best_model.predict_proba(X_test)[: , 1]
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.plot(fpr, tpr, label="ROC Curve")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Best Model")
plt.legend()
plt.show()
```

STEP 14 – Feature Importance (Business

Insight) Use Random Forest for feature importance (even if

XGBoost is best): **Cell 13:**


```
import numpy as np

importances = rf.feature_importances_
indices = np.argsort(importances)[-10:] # Top 10

print("Top 10 important features:")
for i in indices:
    print(X.columns[i], ":", round(importances[i], 4))
```

Then write a Markdown cell:

Key Insights

- Shorter tenure customers are more likely to churn.
- Month-to-month contract shows higher churn.
- Higher MonthlyCharges lead to higher churn.
- Customers without online security / tech support churn more.

(Adapt based on what you actually see.)

STEP 15 – Add a Conclusion Section

Add a final Markdown cell:

Conclusion

- Built churn prediction model using telecom customer data.
- Tried Logistic Regression, Random Forest, and XGBoost.
- Best model achieved ROC-AUC of around XX and recall of YY. - Business can use model to:
 - Give offers to high-risk customers.
 - Focus on long-term contracts.
 - Improve services for high-churn groups.

Replace XX and YY with your scores.

CHAPTER 5 — CONCLUSION

The internship at **AtharavTech Solutions LLP (Drone Technology Division)** proved to be a highly enriching and transformative experience. It offered practical exposure to drone-based data analytics, real time telemetry processing, and business intelligence applications—areas that extend far beyond classroom learning. Working with large datasets, sensor-based flight logs, geospatial information, and analytical dashboards provided deep insights into how data supports automation, monitoring, and decision-making in emerging drone technology industries.

Throughout the internship period, the hands-on tasks enabled meaningful skill development in multiple domains, including **Python programming, SQL querying, Power BI dashboard creation, geospatial mapping, data preprocessing, and exploratory data analysis methodologies**. These activities helped reinforce theoretical concepts through real-world applications and strengthened problem-solving ability while working with dynamic and sometimes noisy datasets.

Overall, this internship successfully fulfilled its intended learning goals. It provided a strong foundation for future work in the domains of **data analytics, automation, artificial intelligence, and drone technology applications**. The knowledge gained, challenges solved, and tools mastered during this period will be instrumental in shaping future academic progress and professional growth. This experience has strengthened interest in pursuing a career in data-driven technology and has encouraged continuous learning in advanced analytics, machine learning, and autonomous system design.

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